BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

RESEARCH / PROTOTYPE SYSTEM DISCLAIMER: This system is a research/prototype fingerprint-based blood-group classification system. It is not clinically validated and must not be used as a substitute for standard serological blood-group testing or for medical/transfusion decisions. All predictions and advisory indicators are provided solely for evaluation purposes.

1. EXAMINATION & SAMPLE AUDIT DETAILS

Report ID:	BG-000565	**Timestamp:**	2026-09-08 19:28:12
Input Source:	Mantra MFS100 Scanner | Standard Finger: LEFT THUMB	**Evaluation Status:**	LOW_CONFIDENCE
Model Name:	Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%)	**Confidence:**	**39.68%**
Standard Acquisition Finger:	**LEFT THUMB** (Convention-based protocol — not automatically verified by scanner)		
Advisory Status:	**NEEDS CONFIRMATION** (τ=70%)	**Advisory Reason:**	**LOW_CONFIDENCE_AND_RH_UNCERTAINTY**

2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP

Predicted Blood Group:

AB+

Confidence Score: **39.68%**

Advisory Threshold: **70%**

Advisory Screening: **NEEDS CONFIRMATION**

Derived ABO Group: **AB** (48.5% prob) | Rh Factor: **+**

RESEARCH ADVISORY: Advisory threshold passed (≥ 70.0%). Note: Confidence OK does not constitute medical verification. Standard serological testing required for medical/transfusion use.

Rh FACTOR UNCERTAINTY ADVISORY: Rh factor (±) determination is uncertain (moderate 8-class confidence (39.7%)). ABO group appears to be 'AB' (48.5% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

A+	A-	AB+	AB-	B+	B-	O+	O-
4.7%	1.4%	39.7%	8.8%	22.0%	2.9%	1.9%	18.7%